

Edufy — Learning Management System

A Java Web Application for streamlined academic management across Admin, Teacher, and Student roles

FINAL YEAR PROJECT

JAVA · JSP · MYSQL · TOMCAT



Project Overview

Domain

Learning Management System (LMS) for educational institutions

Users

Admin, Teachers, Students — each with role-based dashboards

Core Stack

Java Servlet · JSP · JDBC · MySQL · Apache Tomcat

Key Modules

Auth · Subjects · Assignments · Quizzes · Results · Plagiarism Utils



Problem Statement & Proposed Solution

The Problem

- Manual, paper-based assignment & quiz management is slow and error-prone
- No centralized platform for subject content, OTP-based secure login, or performance tracking
- Teachers and students use disconnected tools — email, spreadsheets, WhatsApp
- Zero plagiarism detection or structured result analysis

Edufy's Solution

Unified portal — one login, three role-based dashboards

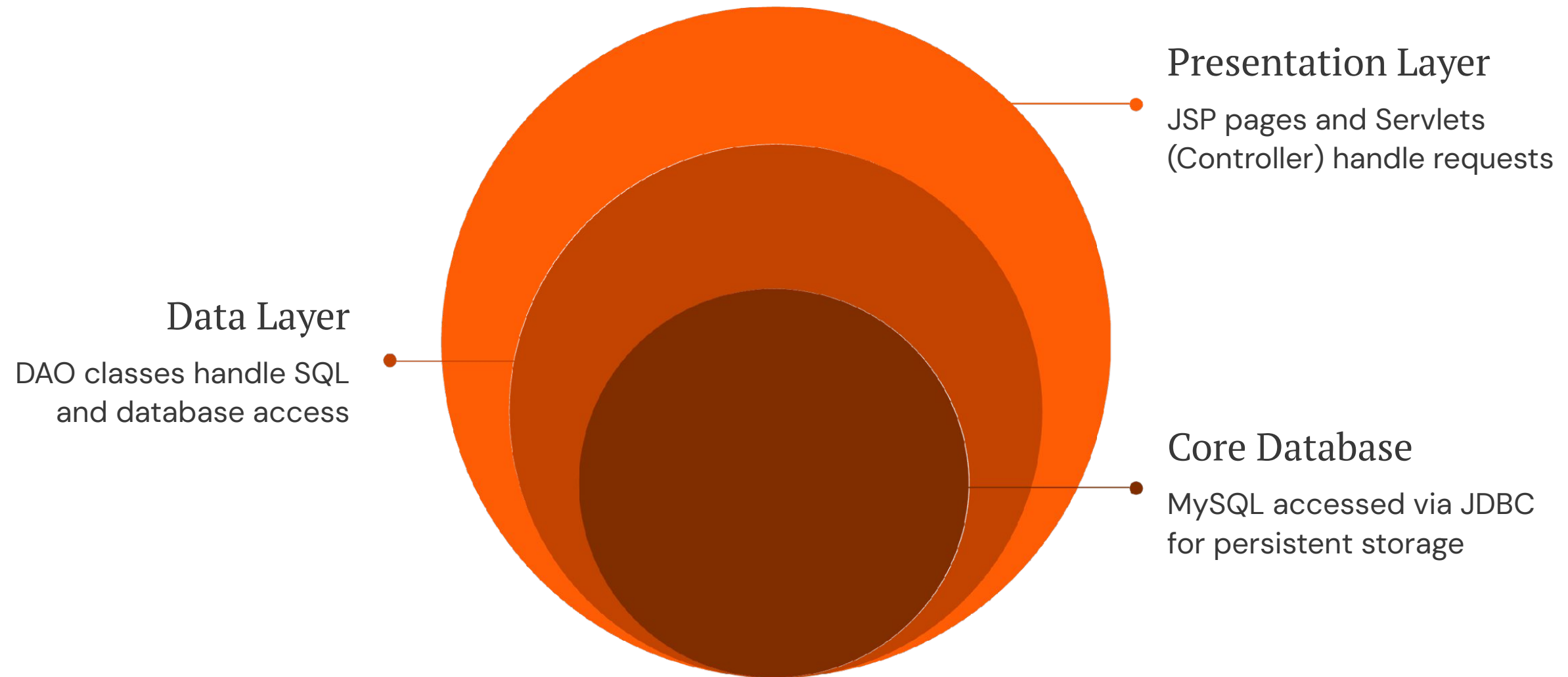
Secure auth — OTP-verified forgot-password + session management

Digital assignments — upload, submit, evaluate all in-app

Quiz engine — attempt, auto-grade, and track results instantly

Keyword/plagiarism utility — basic matching on submissions

System Architecture



Deployed on **Apache Tomcat**. Each HTTP request flows from JSP → Servlet (Controller) → DAO → MySQL, with Model objects carrying data across layers. Utils package handles Email (OTP), Plagiarism, and Keyword matching cross-cutting concerns.

Core Modules — Implementation Map



Authentication

Login · Logout · OTP Forgot Password · Reset Password · Session Guard



Teacher Module

My Subjects · Modules · Quizzes · Upload Assignments · View Submissions · Enter Marks



Quiz Engine

Create questions · Timed attempts · Auto-grading · Result history



Admin Module

Manage Students, Teachers, Subjects · Role assignment · System oversight



Student Module

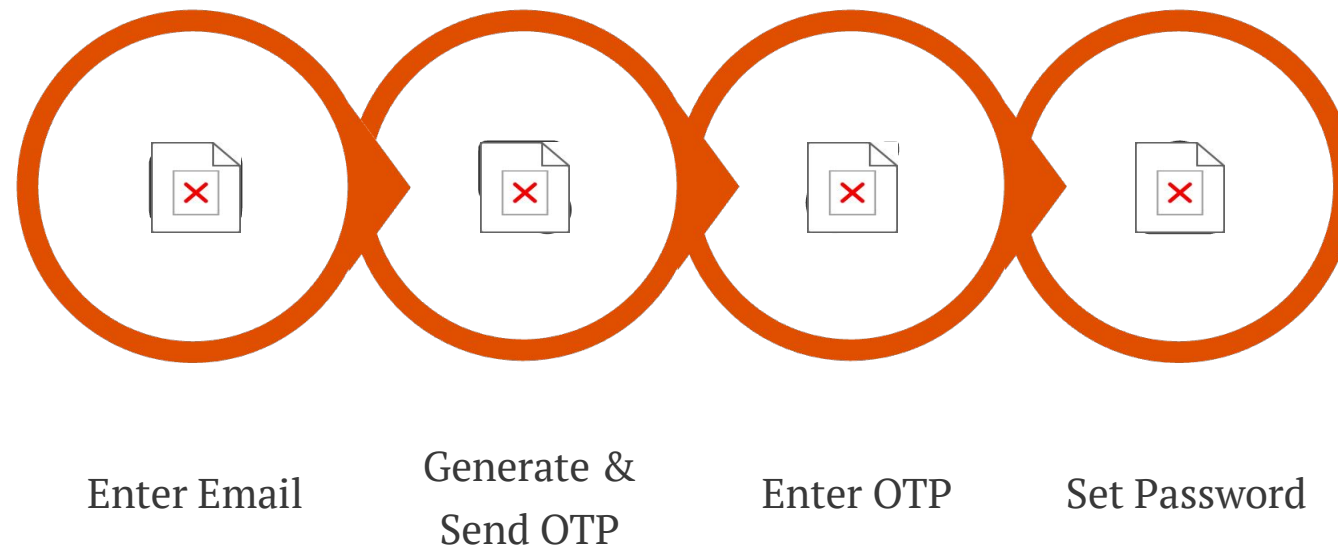
Enroll Subjects · Submit Assignments · Take Quizzes · View Results & Marks



Utilities

EmailUtil (OTP SMTP) · PlagiarismUtil (keyword match) · ValidationUtil

Authentication & OTP Password Reset Flow



OTP is time-limited and single-use. EmailUtil uses JavaMail SMTP to dispatch the code. Failed attempts redirect to error page.

Pages Involved

Login Page

Role detection → session creation

Forgot Password

Email input → OTP trigger

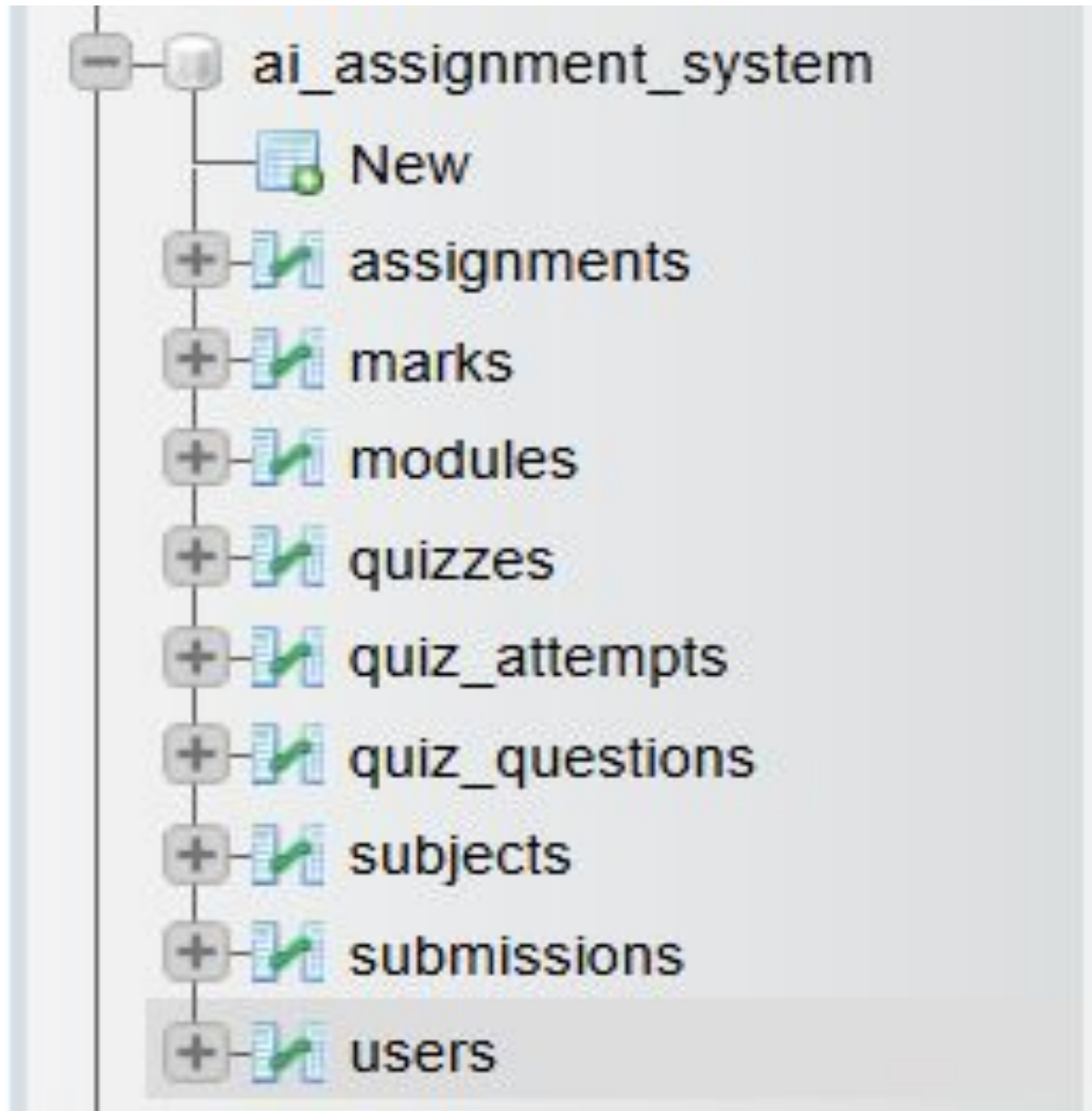
OTP Verify

6-digit code validation

Reset Password

New password + confirm + DB update

Database Design — Key Tables



Design Highlights

users — single table, role column differentiates

Admin/Teacher/Student

enrollments — many-to-many bridge: students ↔ subjects

submissions — stores file path + evaluated marks

quiz_attempts — tracks each student's score per quiz

UI Pages — Role-Based Dashboards



Admin Dashboard

Manage students, teachers, subjects.

Assign roles. View system-wide stats.



Teacher Dashboard

My subjects, upload assignments,
create quizzes, view submissions, enter
marks.



Student Dashboard

Enrolled subjects, submit assignments,
take quizzes, view results and marks.

Testing Strategy & Results

Testing Approach

01

Unit Testing

Individual DAO methods, utility functions (OTP, email, plagiarism logic)

02

Integration Testing

Servlet ↔ DAO ↔ MySQL end-to-end flows for each module

03

System Testing

Complete user journeys — login to quiz result, assignment to mark entry

04

UAT

Verified by simulated admin, teacher, and student role walkthroughs

Sample Test Cases

Test Case	Expected	Status
Login with valid credentials	Redirect to role dashboard	✓ Pass
OTP expiry / wrong OTP	Error message shown	✓ Pass
Student submits assignment	File saved, DB updated	✓ Pass
Quiz auto-grading	Correct score calculated	✓ Pass
Plagiarism keyword match	Match % returned	✓ Pass



Conclusion & Future Scope

What Edufy Delivers

- Fully role-based LMS replacing manual academic workflows
- Secure OTP authentication, structured quiz engine, and digital submissions
- Clean MVC architecture — maintainable and extensible codebase
- Practical proof of Java Servlet + JSP + JDBC + MySQL + Tomcat stack

Future Enhancements



REST API + Mobile App

Android/iOS student app with push notifications



AI-Powered Plagiarism

NLP-based semantic similarity beyond keyword matching

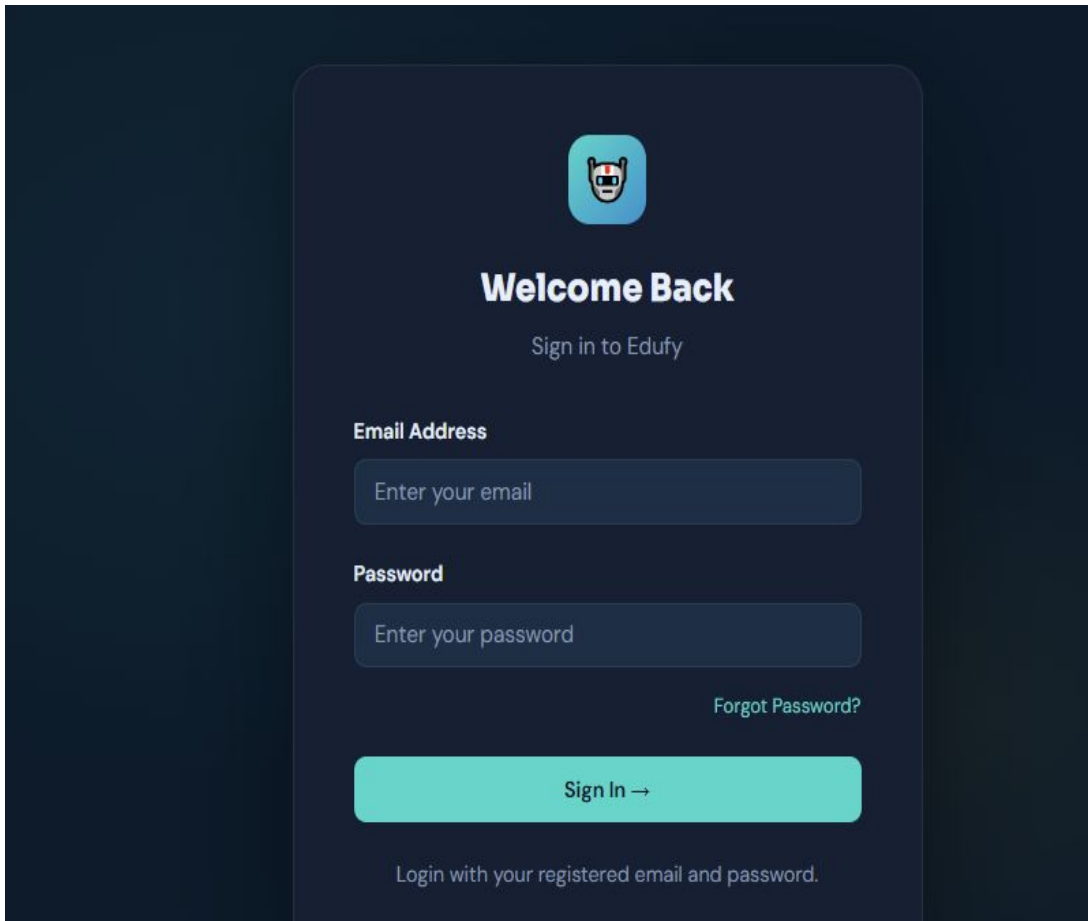


Analytics Dashboard

Charts for attendance, performance trends, and quiz insights

Edufy demonstrates how a structured Java web application can digitize an entire academic ecosystem — from enrollment to evaluation — with security, scalability, and simplicity at its core.

Edufy — Login Page

A screenshot of the Edufy login page. The page has a dark blue background. In the center, there is a white rounded rectangle containing the login form. At the top of this rectangle is a blue square icon with a white robot face. Below the icon, the text "Welcome Back" is written in bold white font, followed by "Sign in to Edufy" in a smaller white font. The form has two input fields: "Email Address" with a placeholder "Enter your email" and "Password" with a placeholder "Enter your password". Below the password field is a link "Forgot Password?" in a small teal font. At the bottom of the form is a teal button with the text "Sign In →". Below the button, there is a small white text: "Login with your registered email and password."

Credential Validation

Username & password verified via DB

Session Management

HttpSession created on success



Backend: **LoginServlet** → **UserDAO.validateUser()** → session
set → role-based redirect

Forgot Password Page



Forgot Password

Enter your registered email to receive OTP.

Email Address

Enter your registered email

Send OTP

← Back to login

Email Lookup

UserDAO checks registered email

OTP Dispatch

EmailService sends OTP to inbox



Backend: **ForgotPasswordServlet** → **UserDAO.findByEmail()** → **EmailService.sendOtp()**

OTP Verification Page

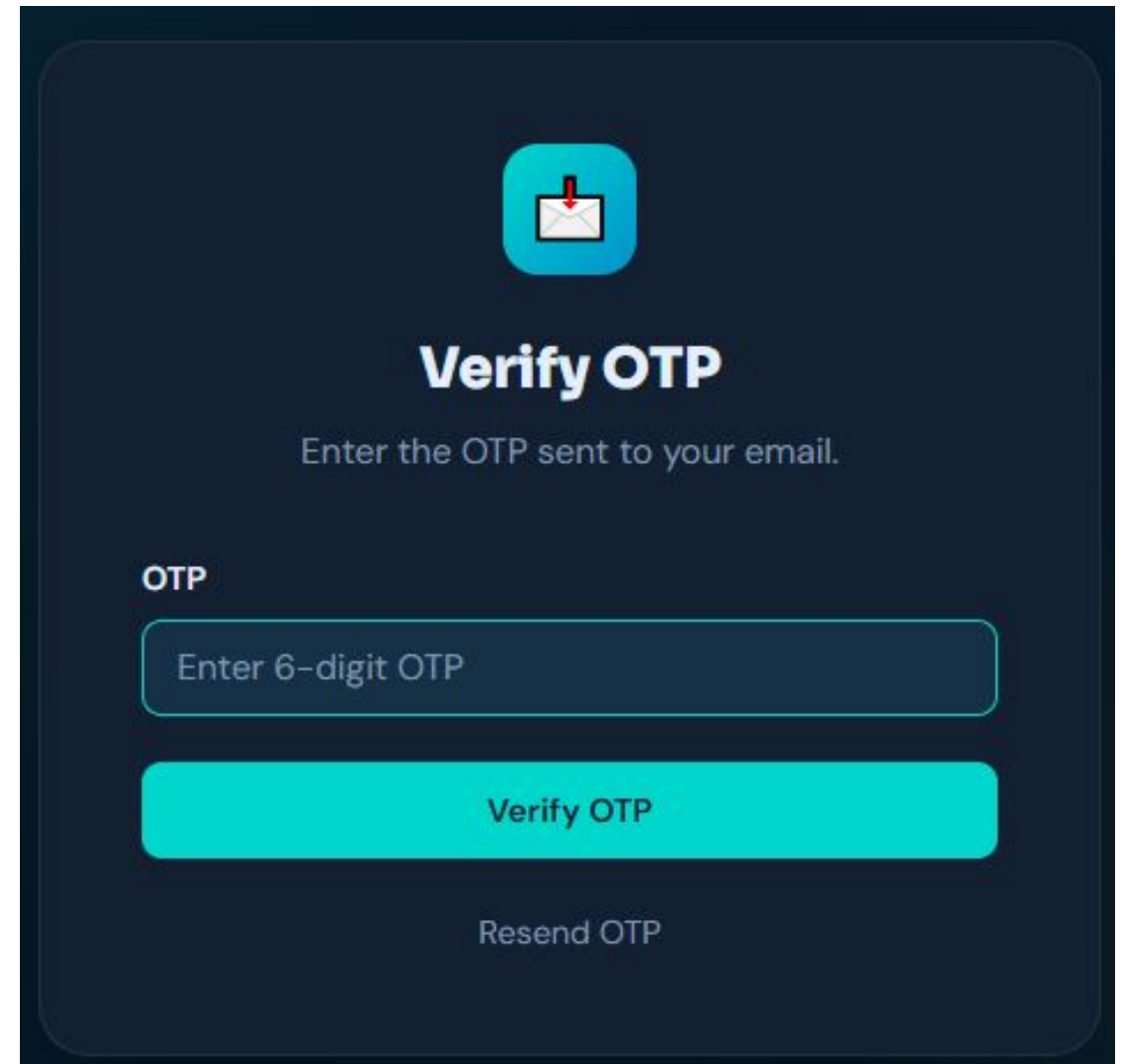
OTP Match Check

Entered OTP compared to stored value

Expiry Validation

OTP invalidated after timeout window

 Backend: **VerifyOtpServlet** → session OTP match → expiry check → forward to reset



The image shows a dark-themed user interface for OTP verification. At the top, there is a blue square icon with a white envelope and a red arrow pointing down. Below this icon, the text "Verify OTP" is displayed in a bold, white font. Underneath, a smaller line of text says "Enter the OTP sent to your email." Further down, the word "OTP" is shown in a light blue font, followed by a light blue rounded rectangular input field containing the placeholder text "Enter 6-digit OTP". Below the input field is a large, solid blue button with the text "Verify OTP" in white. At the bottom of the form, there is a link that says "Resend OTP" in a light blue font.



Reset Password

Create your new password.

New Password

Confirm Password

Reset Password

Reset Password Page

Password Confirmation

Both fields must match exactly

DB Update

Hashed password saved via UserDao



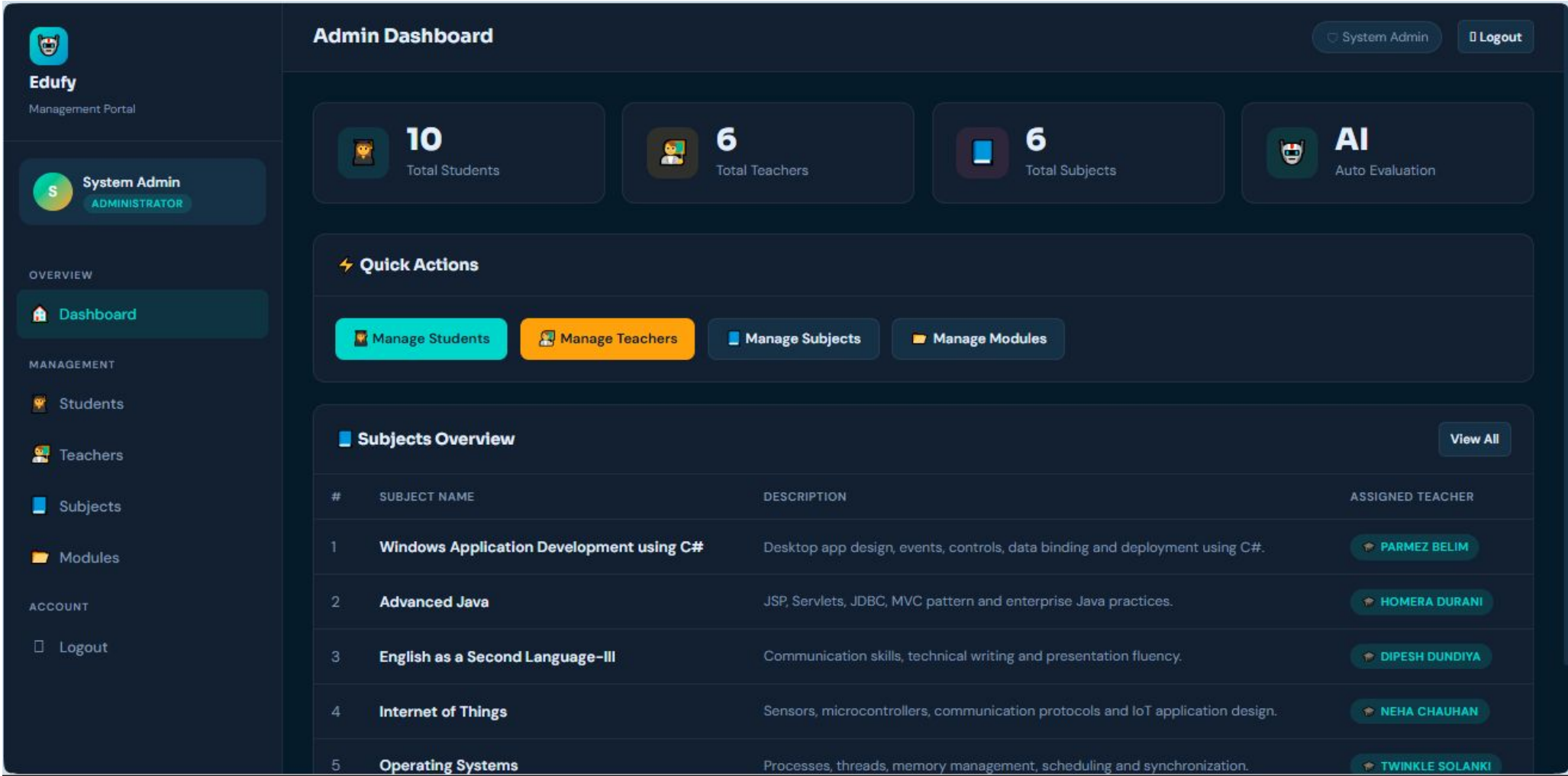
Backend: **ResetPasswordServlet** → validate match →
UserDAO.updatePassword() → redirect login

Admin Dashboard

Role Guard

Admin session verified on load

 Backend: **AdminServlet** → session role check → fetch counts → forward to **adminDashboard.jsp**



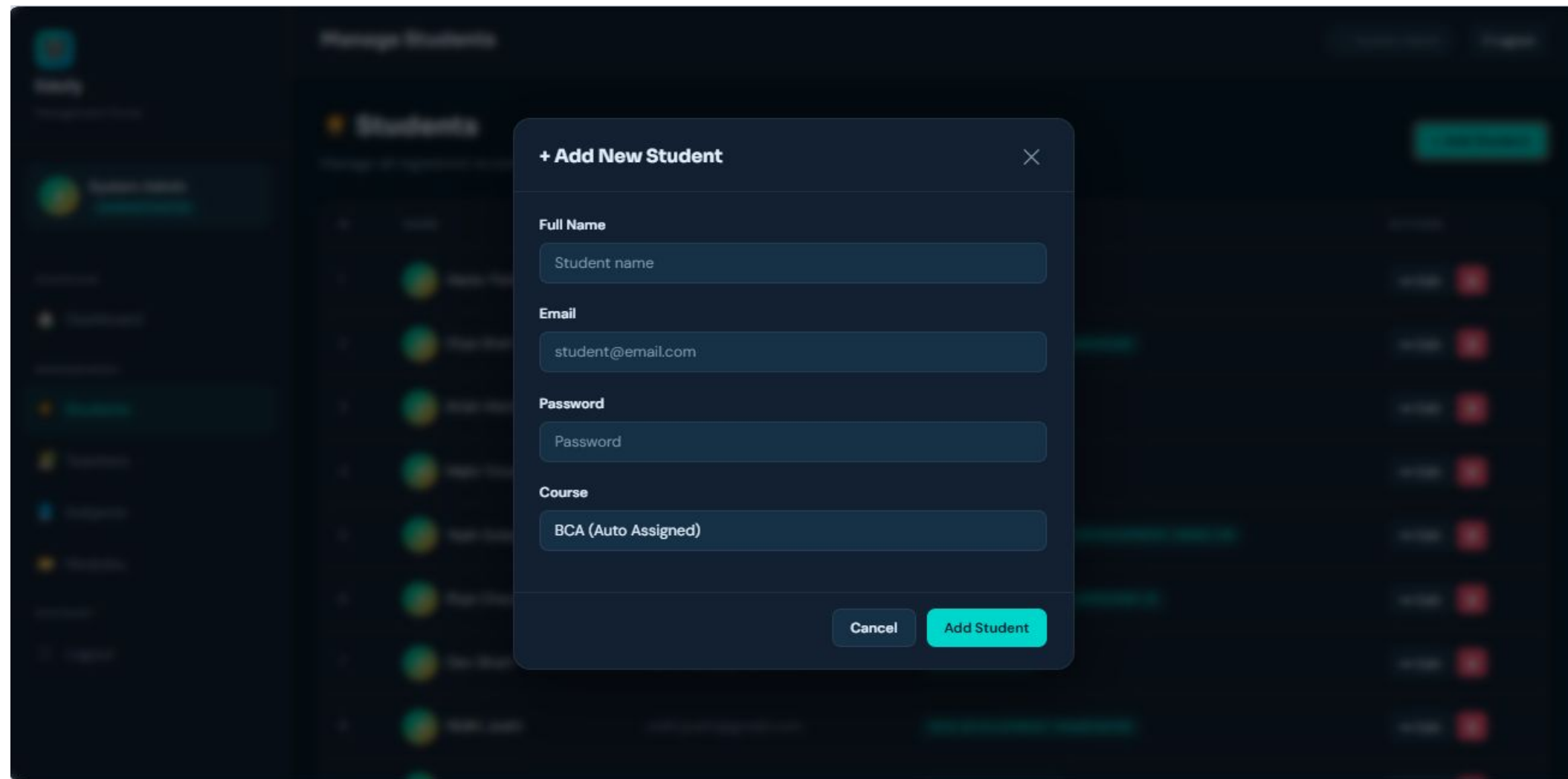
Manage Students

Data Fetch

UserDAO retrieves role=STUDENT rows

CRUD Operations

Add, edit, delete student records



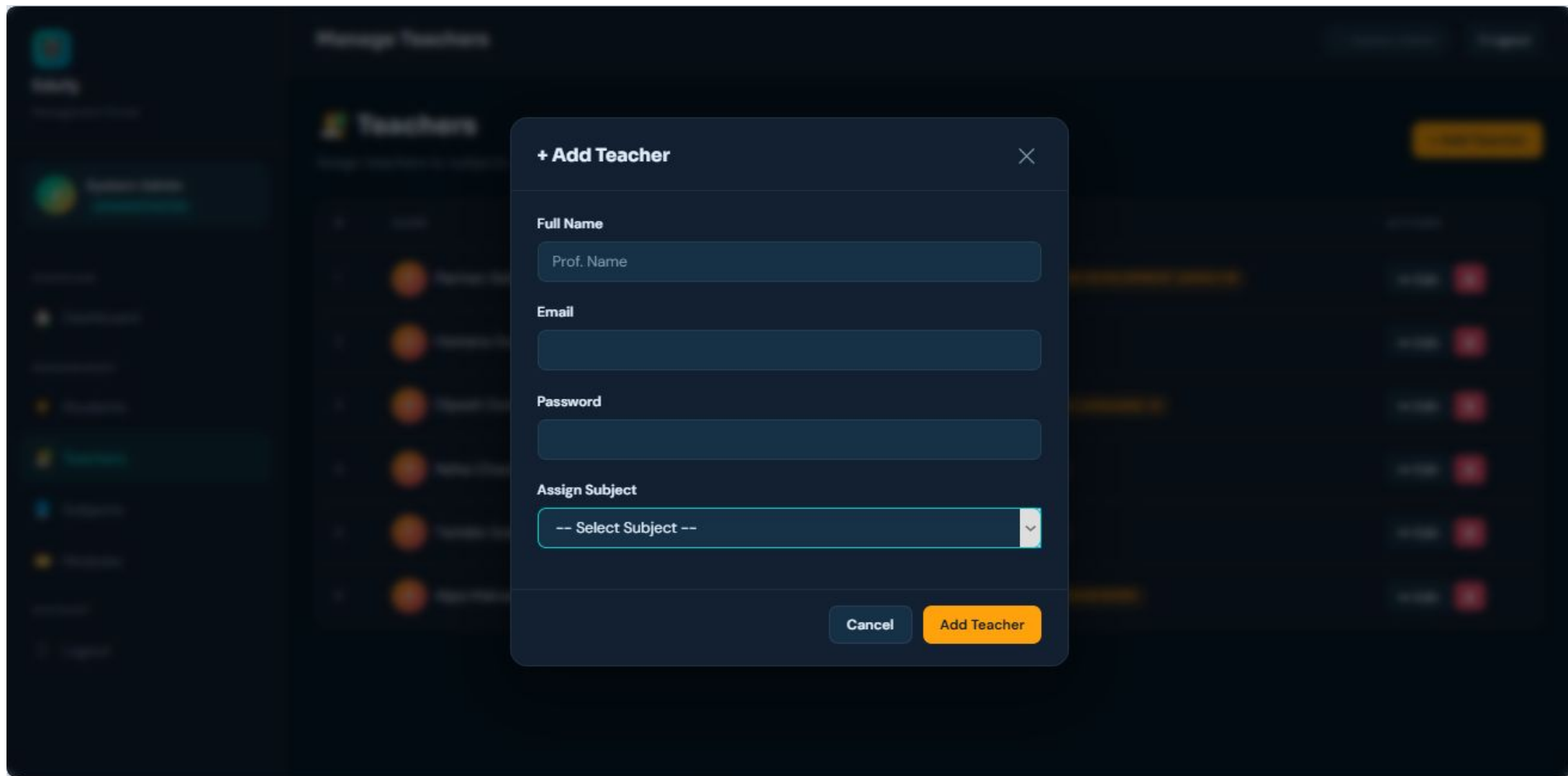
The screenshot displays a web application interface for managing students. A modal window titled '+ Add New Student' is open, featuring a dark theme. The modal contains four input fields: 'Full Name' with the placeholder 'Student name', 'Email' with 'student@email.com', 'Password' with 'Password', and 'Course' with 'BCA (Auto Assigned)'. At the bottom of the modal are two buttons: 'Cancel' and 'Add Student'. The background shows a blurred view of the 'Manage Students' page, which includes a sidebar with navigation options and a main area with a table of student records.

Manage Teachers

Teacher Records

Filter by role=TEACHER in DB

 Backend: `UserDAO getUsersByRole("TEACHER")` → data forwarded to `manageTeachers.jsp`



The screenshot shows a dark-themed web application with a modal window titled "+ Add Teacher" in the center. The modal has a close button (X) in the top right corner. It contains four input fields: "Full Name" (with placeholder text "Prof. Name"), "Email", "Password", and "Assign Subject" (a dropdown menu with "-- Select Subject --"). At the bottom of the modal are two buttons: "Cancel" and "Add Teacher". The background is a blurred view of the "Manage Teachers" page, which shows a list of teachers with columns for name, email, and status.

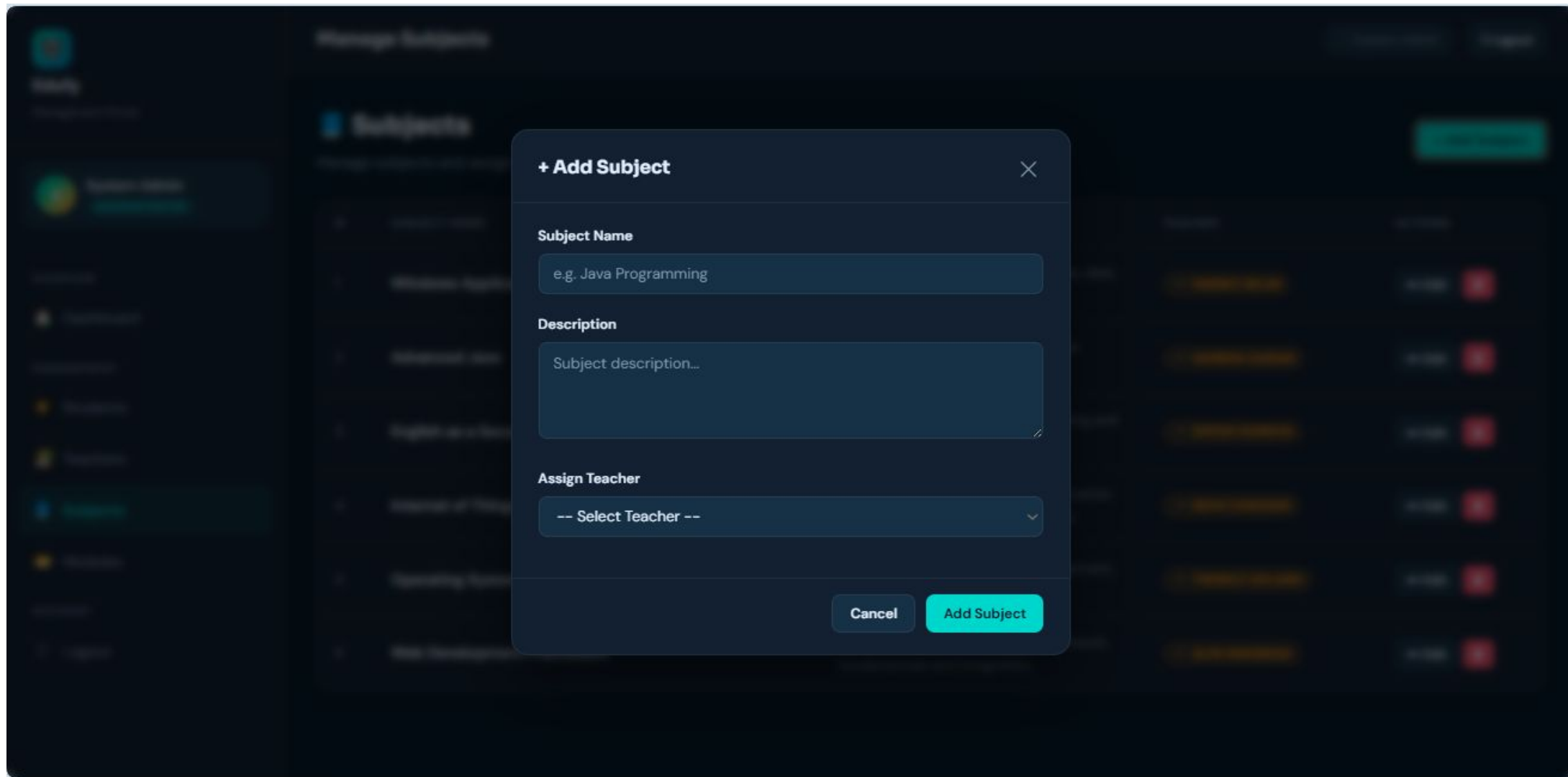
Manage Subjects

DB Layer

SubjectDAO handles all queries



Backend: **SubjectDAO.getAllSubjects()** / **.addSubject()** / **.deleteSubject()** → **manageSubjects.jsp**

A screenshot of a web application interface with a dark theme. In the center, a modal window titled '+ Add Subject' is open. The modal contains three input fields: 'Subject Name' with the placeholder text 'e.g. Java Programming', 'Description' with the placeholder text 'Subject description...', and 'Assign Teacher' which is a dropdown menu currently showing '-- Select Teacher --'. At the bottom of the modal are two buttons: 'Cancel' and 'Add Subject'. The background shows a blurred view of a 'Manage Subjects' page with a table of subjects and a sidebar on the left.

Teacher-Facing LMS Interface

A walkthrough of the core teacher dashboard features — from subject management to submission review and plagiarism detection.

BACKEND-INTEGRATED UI REVIEW



Teacher Dashboard

Overview Panels

At-a-glance summaries of subjects, upcoming deadlines, and student activity

 **Backend:** TeacherServlet

**Edufy**
Teacher Portal

**Homera Durani**
TEACHER

OVERVIEW

Dashboard

TEACHING

My Subjects

Manage Modules

Assignments

Submissions

Marks

ACCOUNT

Logout

Teacher Dashboard

Homera Durani

Logout

**1**
My Subjects

**24**
Assignments Created

**Auto**
Keyword Evaluation

**AI**
Plagiarism Detection

**12**
Quizzes Created

Quick Actions

Manage Modules

+ Upload Assignment

+ Create Quiz

View Submissions

View Marks

My Subjects

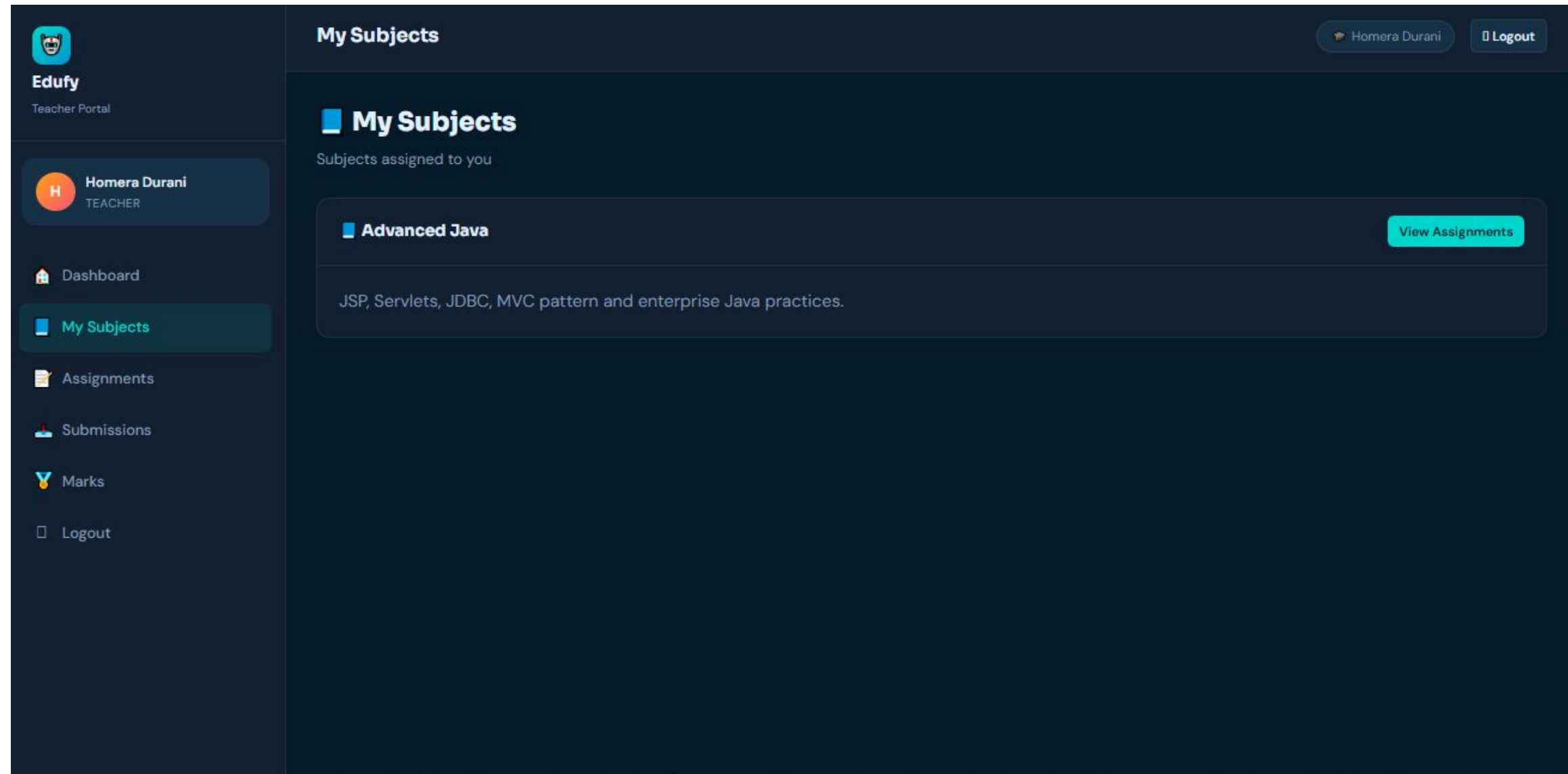
SUBJECT	DESCRIPTION	ACTIONS
Advanced Java	JSP, Servlets, JDBC, MVC pattern and enterprise Java practices.	View Assignments

Teacher My Subjects

Subject List View
Manage Subjects



Backend: TeacherServlet, SubjectDAO

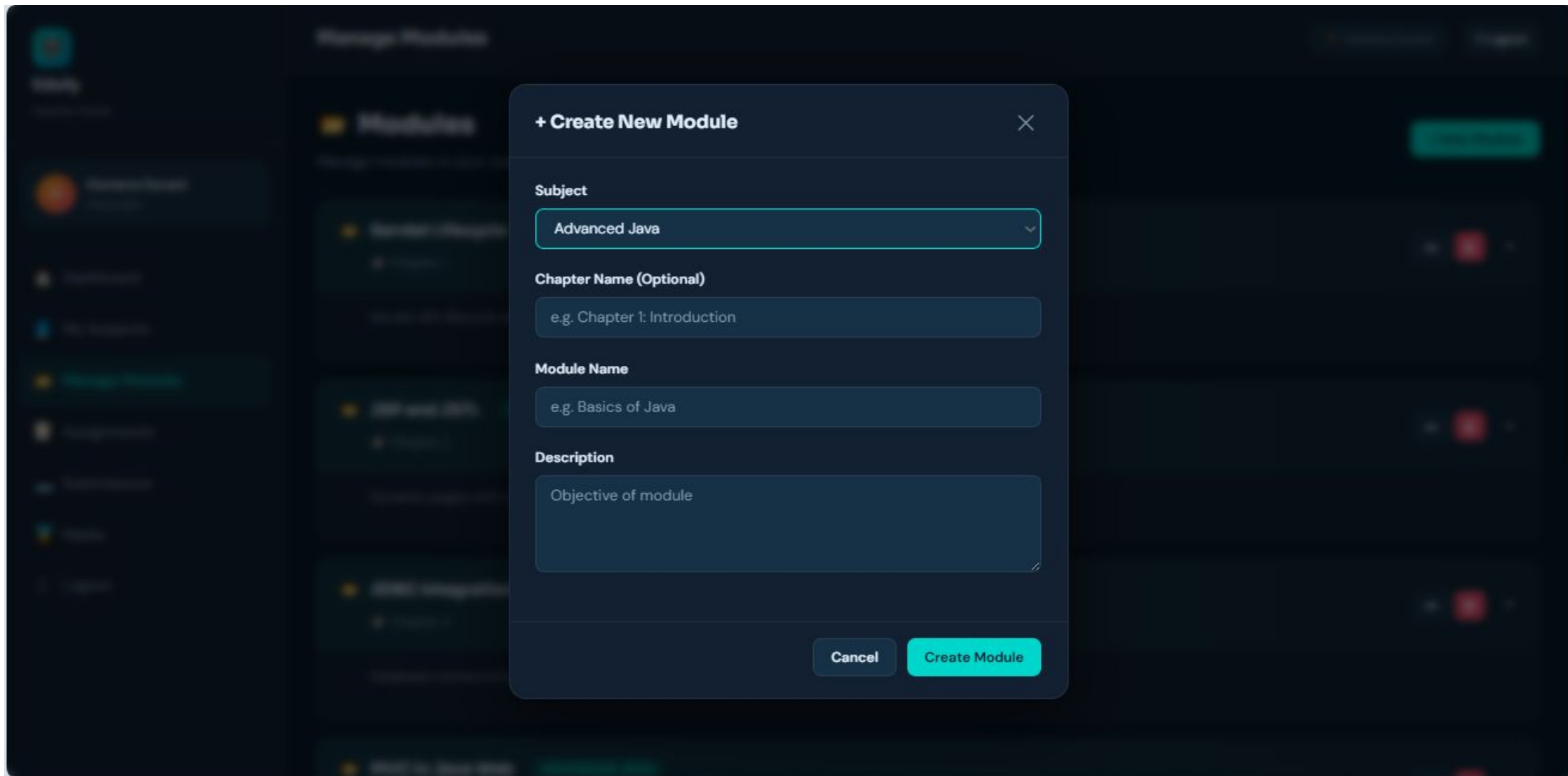


Manage Modules

Add / Edit Modules

Structured Content

📱 **Backend:** ModulesApiServlet, ModuleDAO



The screenshot shows a dark-themed application interface with a modal dialog titled "+ Create New Module". The dialog contains the following fields:

- Subject:** A dropdown menu with "Advanced Java" selected.
- Chapter Name (Optional):** A text input field containing "e.g. Chapter 1: Introduction".
- Module Name:** A text input field containing "e.g. Basics of Java".
- Description:** A text area containing "Objective of module".

At the bottom of the dialog are two buttons: "Cancel" and "Create Module". The background shows a blurred sidebar with navigation icons and a list of modules.

+ Create New Quiz

Module

-- Select Module --

Time Limit (minutes)

ID

Quiz Title

Description

MCQ Questions

+ Add Question

Question 1

Remove

Question

Option A

Option B

Option C

Option D

Correct Answer

A

Marks

1

Cancel

Create Quiz

Manage Quizzes

Build and manage assessments directly within the platform. Teachers retain full control over question type, order, and content — enabling targeted, curriculum-aligned evaluation.



Create Quizzes

Define quiz title, instructions, and time parameters for each assessment



Question Control

Add, edit, reorder, and remove individual questions with answer options



Backend: QuizServlet, QuizDAO

+ New Assignment

×

Module

-- Select Module --

Only modules from your admin-assigned subjects are shown.

Assignment Title

e.g. Java Variables Assignment

Question / Description

Describe what students need to do...

Keywords (comma-separated, used for auto-evaluation)

variable, datatype, int, String, boolean

⚡ Each keyword found in student answer contributes proportional marks

Submission Method

Text Answer

Max Marks

10

Cancel

Create Assignment

Upload Assignment

File Upload

Teachers publish assignment briefs by uploading supporting files — instructions, rubrics, or reference documents — directly to the relevant subject.

Deadline Setting

Each assignment includes a configurable due date and time, automatically surfaced to students in their view.



Backend: SubmissionServlet, AssignmentDAO

View Submissions

Backend: SubmissionServlet, SubmissionDAO

**Edufy**
Teacher Portal

H

Homera Durani
TEACHER

Dashboard

My Subjects

Manage Modules

Assignments

Submissions

Marks

Logout

View Submissions

Homera DuraniLogout

 **Submissions**

Student submissions for your assignments

JSP and JSTL - Theory Assignment (Advance

View Submissions

Submissions — JSP and JSTL - Theory Assignment

3 TOTAL

#	STUDENT	SUBMITTED AT	STATUS	FILE	CONTENT PREVIEW
1	A Aarav Patel	2026-03-28 02:40:48	EVALUATED	No file	Submission by student Aarav Patel for assignment JSP and JSTL - Theory Assignmen...
2	M Mahi Trivedi	2026-03-22 02:40:48	PENDING	No file	Submission by student Mahi Trivedi for assignment JSP and JSTL - Theory Assignme...
3	D Dev Bhatt	2026-03-16 02:40:48	PENDING	No file	Submission by student Dev Bhatt for assignment JSP and JSTL - Theory Assignment...

Plagiarism & Keyword Results

 Backend: PlagiarismChecker, KeywordMatcher



Edufy

Teacher Portal

Homera Durani

TEACHER

Dashboard

My Subjects

Manage Modules

Assignments

Submissions

Marks

Logout

View Marks

Homera Durani

Logout

Student Marks

Auto-evaluated marks based on keyword matching

Servlet Lifecycle – Theory Assignment (Advance

View Marks

ASSIGNMENT

MAX MARKS

KEYWORDS

Servlet Lifecycle – Theory Assignment 10

SERVLET

LIFECYCLE

CONCEPT

EXAMPLE

IMPLEMENTATION

USECASE

Marks Report

1 EVALUATED

#	STUDENT	MARKS	SCORE %	MATCHED KEYWORDS	PLAGIARISM	EVALUATED AT
1	Riya Chauhan	7 / 10	70%	CONCEPT EXAMPLE IMPLEMENTATION	20% SIMILAR	2026-04-01 02:40:49

System Feedback States

Clear, consistent UI feedback is essential for trust. The interface handles both success and failure gracefully — confirming completed actions and guiding users through error or empty states.

Success Message UI

Confirmation banners appear after key actions — submitting, saving, or uploading — so teachers know the system responded correctly.

Backend: Servlet response messaging

Error / Empty State

When no data is available or an action fails, the interface displays a clear message rather than a blank screen — preserving usability and reducing confusion.

Backend: error.jsp, DAO no-data handling

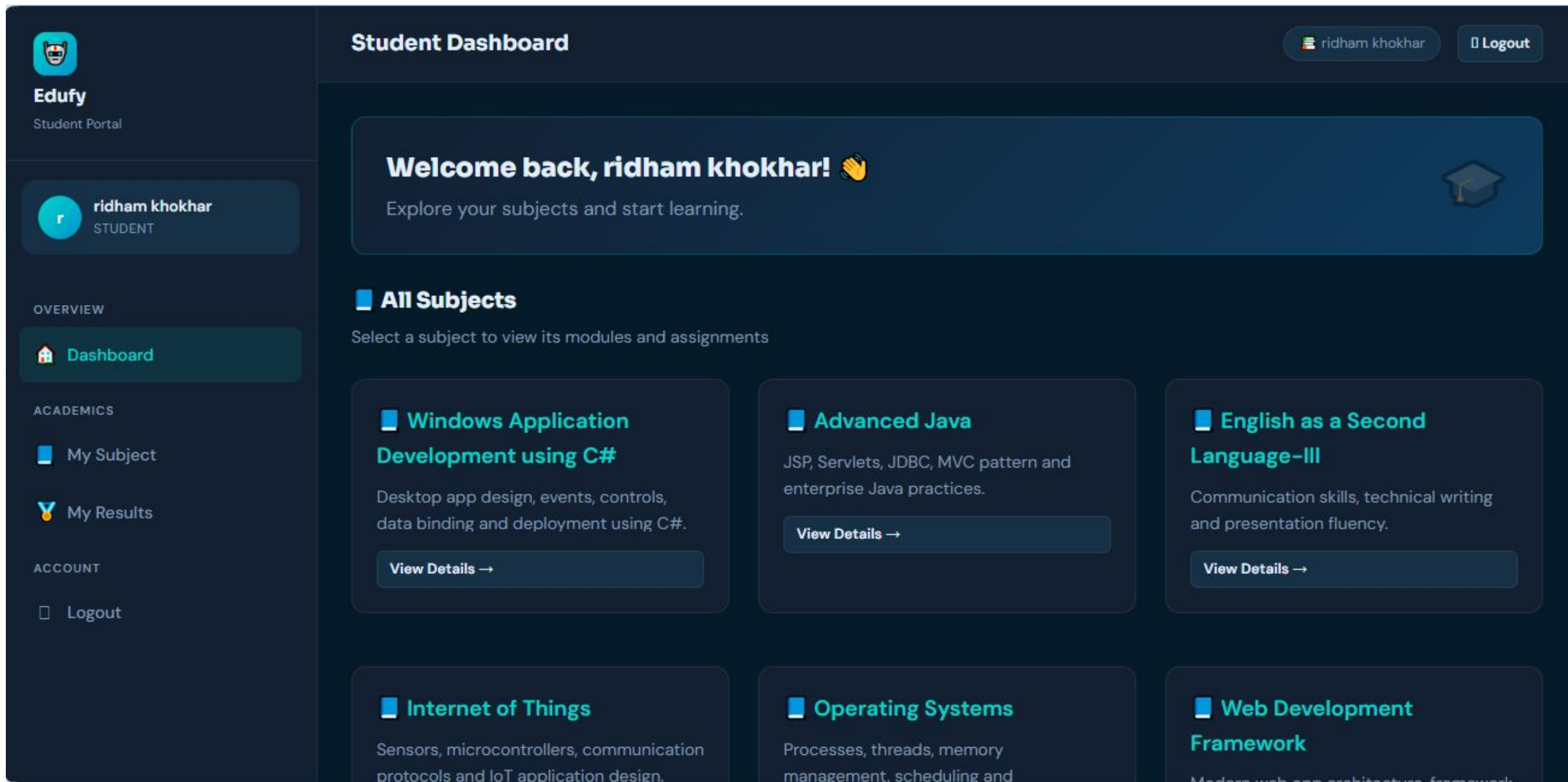
Edufy LMS — Student Dashboard

Backend Mapping

Servlet: StudentServlet.java

Action: loadDashboard()

Session: student_id validated



The screenshot displays the Edufy LMS Student Dashboard. The interface is dark-themed with a sidebar on the left and a main content area. The sidebar includes the Edufy logo, the user's name 'ridham khokhar' with a profile picture, and navigation links for 'OVERVIEW' (Dashboard), 'ACADEMICS' (My Subject, My Results), and 'ACCOUNT' (Logout). The main content area is titled 'Student Dashboard' and shows a welcome message for 'ridham khokhar!'. Below this, there is a section for 'All Subjects' with a list of subjects: 'Windows Application Development using C#' (Desktop app design, events, controls, data binding and deployment using C#), 'Advanced Java' (JSP, Servlets, JDBC, MVC pattern and enterprise Java practices), 'English as a Second Language-III' (Communication skills, technical writing and presentation fluency), 'Internet of Things' (Sensors, microcontrollers, communication protocols and IoT application design), 'Operating Systems' (Processes, threads, memory management, scheduling and), and 'Web Development Framework' (Modern web app architecture framework). Each subject card has a 'View Details' button.

Student Dashboard

Welcome back, ridham khokhar! 🙌

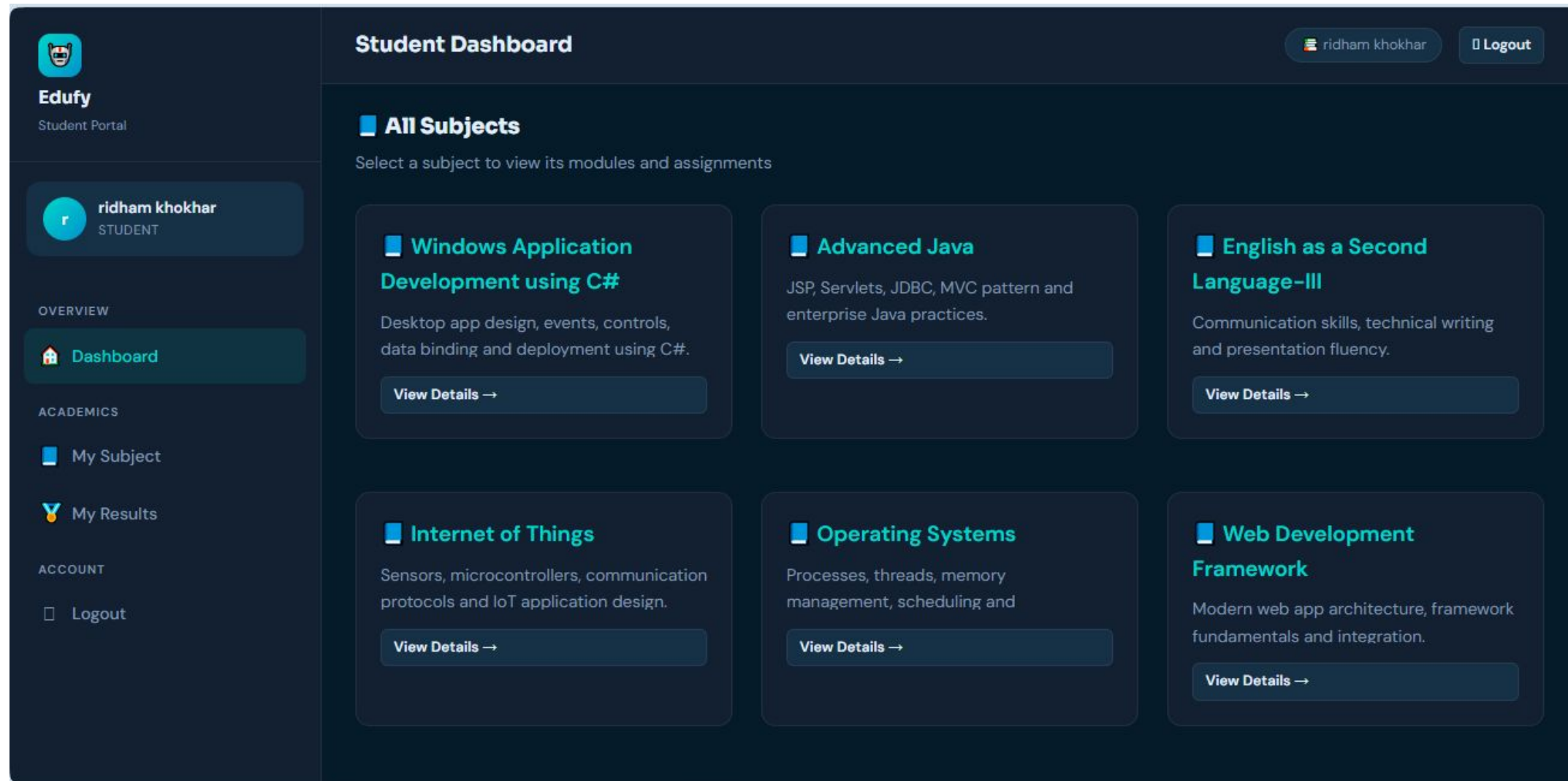
Explore your subjects and start learning.

All Subjects

Select a subject to view its modules and assignments

- Windows Application Development using C#**
Desktop app design, events, controls, data binding and deployment using C#.
[View Details →](#)
- Advanced Java**
JSP, Servlets, JDBC, MVC pattern and enterprise Java practices.
[View Details →](#)
- English as a Second Language-III**
Communication skills, technical writing and presentation fluency.
[View Details →](#)
- Internet of Things**
Sensors, microcontrollers, communication protocols and IoT application design.
- Operating Systems**
Processes, threads, memory management, scheduling and
- Web Development Framework**
Modern web app architecture framework.

My Subjects



Backend Mapping

Servlet: StudentServlet.java

DAO:

SubjectDAO.getSubjectsByStudent(id)

DB Table: enrollments, subjects

Subject Details

MODULEDAO

ASSIGNMENTDAO

QUIZDAO

Backend Mapping

DAOs: ModuleDAO,
AssignmentDAO, QuizDAO

Method: getBySubject(subjectId)

View: subjectDetail.jsp

The screenshot displays the Edufy Student Portal interface. On the left is a dark sidebar with the Edufy logo and 'Student Portal' text. Below this, a user profile for 'ridham khokhar' (STUDENT) is shown. The sidebar menu includes 'Dashboard', 'My Subject' (highlighted), 'My Results', and 'Logout'. The main content area is titled 'Subject Details' and features a user profile 'ridham khokhar' with a 'Logout' button. The subject 'Advanced Java' is highlighted, with a description: 'JSP, Servlets, JDBC, MVC pattern and enterprise Java practices.' Below this, 'Chapter 1' is expanded to show 'Servlet Lifecycle' with '4 ASSIGNMENTS'. Under 'Module Quizzes', there are two quiz entries: 'Servlet Lifecycle Advanced Quiz' (5 MCQ | 25 min) and 'Servlet Lifecycle Quiz' (5 MCQ | 20 min), each with a 'Start Quiz' button. At the bottom, there are two assignment entries: 'Servlet Lifecycle - Theory Assignment' (10 MARKS) and 'Servlet Lifecycle - Practical Assignment' (20 MARKS), each with a 'Submit' button. A question prompt is visible under the theory assignment: 'Question: Explain key concepts of Servlet Lifecycle with suitable practical examples.'

Submit Assignment

SUBMISSIONSERVLET

SUBMISSIONDAO

Backend Mapping

Servlet: SubmissionServlet.java

DAO:

SubmissionDAO.save(submission)

Storage: /uploads/ directory

The screenshot shows the 'Submit Assignment' page in the Edufy Student Portal. The page has a dark blue theme. On the left is a sidebar with the user's name 'ridham khokhar' and 'STUDENT', and navigation links for 'Dashboard', 'My Subject', 'My Results', and 'Logout'. The main content area is titled 'Submit Assignment' and shows the assignment details for 'Servlet Lifecycle - Theory Assignment - Advanced Java'. It includes a description, a tip about keywords, and a large text area for the answer. At the bottom, there are 'Submit Assignment' and 'Cancel' buttons. The user's name and a 'Logout' link are in the top right corner.

Submit Assignment

Servlet Lifecycle - Theory Assignment - Advanced Java

Servlet Lifecycle - Theory Assignment

Explain key concepts of Servlet Lifecycle with suitable practical examples.

10 MAX MARKS **SERVLET LIFECYCLE**

Tip: Make sure your answer includes relevant keywords for better marks. Answers are auto-evaluated.

Your Submission

ALLOWED SUBMISSION METHOD: TEXT

Type or paste your answer below

Write your complete answer here. Include relevant keywords from the assignment question for better evaluation.

Words: 0

Submit Assignment **Cancel**

Take Quiz — Start Screen

QUIZSERVLET

Backend Mapping

Servlet: QuizServlet.java

Action: action=startQuiz

Session: quiz_id, student_id
stored

The screenshot shows the 'Take Quiz' interface in the Edify Student Portal. The user is logged in as 'ridham khokhar' (STUDENT). The main heading is '? Servlet Lifecycle Advanced Quiz' with a subtitle 'Scenario-based and application-level questions for Servlet Lifecycle'. The interface displays three questions (Q1, Q2, Q3) with multiple-choice options. A 'Countdown' timer shows 24:53. A 'Progress' bar indicates 'Filled 0 of 5 MCQ'. The left sidebar contains navigation links: Dashboard, My Results, and Logout.

Take Quiz

? Servlet Lifecycle Advanced Quiz
Scenario-based and application-level questions for Servlet Lifecycle

Q1. Which option best describes core concept of Servlet Lifecycle Advanced Quiz?

- ☐ A. Core Concept
- ☐ B. Optional Detail
- ☐ C. Irrelevant Point
- ☐ D. None

Marks: 1

Q2. Which practice is recommended while working on Servlet Lifecycle Advanced Quiz?

- ☐ A. Ignore validation
- ☒ B. Follow standards
- ☐ C. Skip testing
- ☐ D. Avoid documentation

Marks: 1

Q3. What Improves quality in Servlet Lifecycle Advanced Quiz?

- ☐ A. No review

Countdown
24:53
Time runs from first click/start.

Progress
Filled 0 of 5 MCQ

Edify
Student Portal

ridham khokhar
STUDENT

[Dashboard](#)

[My Results](#)

[Logout](#)

Take Quiz — Question Attempt

QUIZDAO

QUIZATTEMPT LOGIC

Backend Mapping

DAO:

QuizDAO.getQuestions(quizId)

Logic: QuizAttempt.java (POJO)

Session: answers[] stored per question

The screenshot displays the Edify Student Portal interface during a quiz attempt. The left sidebar shows the user's profile (ridham khokhar, STUDENT) and navigation links (Dashboard, My Results, Logout). The main content area is titled "Take Quiz" and shows the "Servlet Lifecycle Advanced Quiz". The quiz description is "Scenario-based and application-level questions for Servlet Lifecycle". The interface displays three questions (Q1, Q2, Q3) with multiple-choice options. Q1 asks for the core concept of Servlet Lifecycle Advanced Quiz, with options: A. Core Concept, B. Optional Detail, C. Irrelevant Point, D. None. Q2 asks for the recommended practice while working on Servlet Lifecycle Advanced Quiz, with options: A. Ignore validation, B. Follow standards, C. Skip testing, D. Avoid documentation. Q3 asks what improves quality in Servlet Lifecycle Advanced Quiz, with option A. No review. The right sidebar shows a countdown timer (23:50) and progress (Filled 3 of 5 MCQ).

Take Quiz

? Servlet Lifecycle Advanced Quiz
Scenario-based and application-level questions for Servlet Lifecycle

Q1. Which option best describes core concept of Servlet Lifecycle Advanced Quiz?

- ☐ A. Core Concept
- ☐ B. Optional Detail
- ☐ C. Irrelevant Point
- ☐ D. None

Marks: 1

Q2. Which practice is recommended while working on Servlet Lifecycle Advanced Quiz?

- ☐ A. Ignore validation
- ☐ B. Follow standards
- ☐ C. Skip testing
- ☐ D. Avoid documentation

Marks: 1

Q3. What improves quality in Servlet Lifecycle Advanced Quiz?

- ☐ A. No review

Countdown
23:50
Time runs from first click/start.

Progress
Filled 3 of 5 MCQ

Quiz Submit Confirmation

QUIZSERVLET FLOW

Backend Mapping

Servlet: QuizServlet.java

Action: action=submitQuiz

Flow: evaluate → persist → redirect

The screenshot displays the Edufy Student Portal interface. On the left is a dark sidebar with the Edufy logo and navigation links: Dashboard, My Subject (highlighted), My Results, and Logout. The main content area is titled 'Subject Details' and shows a confirmation message: 'Quiz submitted. Score: 3/5'. Below this, the subject 'Advanced Java' is listed with its topics. Under 'Chapter 1', the 'Servlet Lifecycle' section is expanded, showing '4 ASSIGNMENTS'. It includes two quizzes: 'Servlet Lifecycle Advanced Quiz' (5 MCQ | 25 min, Score: 1/5) and 'Servlet Lifecycle Quiz' (5 MCQ | 20 min, Score: 3/5). Both are marked 'SUBMITTED'. Below the quizzes are two assignments: 'Servlet Lifecycle - Theory Assignment' (10 MARKS) and 'Servlet Lifecycle - Practical Assignment' (20 MARKS), each with a 'Submit' button. The question for the theory assignment is 'Explain key concepts of Servlet Lifecycle with suitable practical examples.' and for the practical assignment is 'Build a mini practical task for Servlet Lifecycle and submit report/screenshot/file.'

Quiz submitted. Score: 1/5

View Result

MARKSDAO

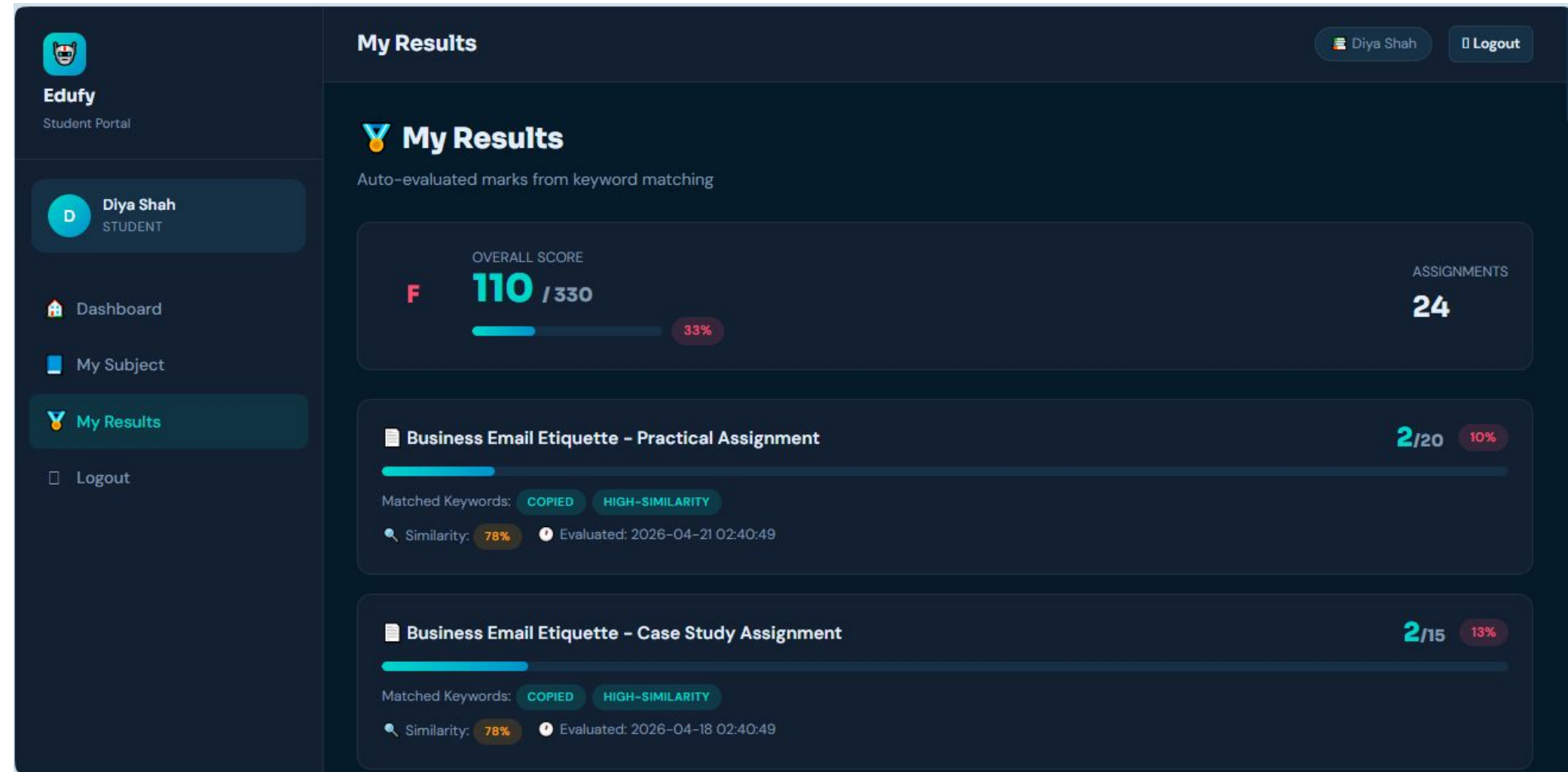
Backend Mapping

DAO:

MarksDAO.getMarksByStudent(id)

DB Table: marks, quizzes,
assignments

View: result.jsp



Thank You!

Ridham Khokhar

BCA Semester 4